Batching, mixing, transporting, placing and compaction

Notes from the lecture slides "Module 5 – Mixing, Placing and Compaction of concrete (Batching, Mixing and placing of concrete)"

In this module

1. Production of concrete
2. Batching
3. Mixing
4. Transporting concrete
5. Placing concrete
6. Compaction
7. Defects from improper compaction

1 Production of concrete

Making concrete on site is a chain of operations; a mistake at any stage spoils the final product.



Stages in producing concrete. (Curing is covered with fresh and hardened concrete.)

2 Batching

Batching is **measuring the specified quantities** of cement, aggregates, water and admixtures for each batch.

2.1 Methods

Volume batching	Weigh batching
Materials measured by volume (gauge boxes, buckets)	Materials measured by **mass** on scales
Simple and cheap, for small, unimportant work	**Accurate**; required by **IS 456:2000** for all important work
Inaccurate: bulking of moist sand, loose vs compacted filling	Unaffected by bulking; moisture still corrected for

2.2 Storage of materials

- **Aggregate bins** store the different aggregate sizes separately.
- **Silos** store cement and other cementitious materials (fly ash, GGBS), keeping them dry.

2.3 Components of a batching plant

- **Aggregate bins** for each type and size of aggregate
- **Feeding mechanisms** (scrapers, conveyors, hoists) to move aggregate to the scales
- **Weighing and measuring system**: mechanical lever systems or **load cells**; electrical or **microwave moisture gauges** to measure aggregate moisture (so water can be corrected automatically)
- **Cement silos** with a **screw conveyor** or bucket conveyor
- **Water tank** and water-measuring system
- **Admixture dispenser**

2.4 Batching tolerances

Ingredient	Tolerance
Cement and other cementitious materials	**± 1%**
Water (by volume or weight)	**± 1%**
Aggregates	**± 2%**
Admixtures (by volume or weight)	**± 3%**

Accurate batching gives **uniform concrete with less variation**, so a lower margin (standard deviation) is needed and the mix is more economical.

3 Mixing

Mixing coats the surface of all aggregate particles with cement paste and blends all ingredients into a **uniform mass**, by rotation or stirring. This uniformity must **not be lost while discharging** the concrete from the mixer.

Two methods:

1. **Hand mixing**, by labour: for small work. (Usually 10% extra cement is added to make up for poorer mixing.)
2. **Machine mixing**: for larger work, and in batching plants where continuous, automated mixing is needed.

3.1 Types of machine mixers

- **Batch mixers**: one batch is mixed and **fully discharged before** the next is loaded.
- **Continuous mixers**: for very large projects.

There are four types of batch mixer.

Mixer	How it works	Points to remember
Tilting drum	Conical or bowl-shaped drum with internal vanes; **tilted to discharge**	**Rapid, unsegregated discharge**. Suitable for **low-workability mixes** and **large aggregate**
Non-tilting drum	Axis always **horizontal**; discharged by inserting a **chute** into the drum or by **reversing** the drum	**Slow discharge**; some **segregation** (part of the coarse aggregate comes out last). Loaded by a **skip**; the whole charge must go in each time
Pan-type (forced action)	A circular **pan rotates**, with one or two sets ("stars") of **paddles** rotating about a vertical axis that is not the pan's axis. Paddle height can be adjusted to stop mortar coating the pan bottom	Mixes by **forced action**, not free fall. Effective for **stiff, cohesive mixes**; used in **precast** works and **laboratories**

Mixer	How it works	Points to remember
Dual drum	**Two drums in series**: partly mixed in the first, transferred to the second to finish. Meanwhile the first is recharged	Used in **highway construction**. Batches do not intermix; **output is doubled**

Continuous mixers: loading, mixing and discharging go on **continuously** until work stops. Materials are fed continuously by **screw feeders**. Used for very large projects: **dams, bridges, high-rise buildings**.

3.2 Uniformity of mixing (ASTM C94)

Two samples taken from different parts of a batch (e.g. from the first and last portions discharged) should not differ by more than:

Property	Maximum difference
Density of concrete	16 kg/m^3
Air content	1%
Slump (average below 100 mm)	25 mm
Slump (average 100–150 mm)	40 mm
Coarse aggregate (retained on 4.75 mm sieve)	6%
Density of air-free mortar	1.6%
Compressive strength (average 7-day of 3 cylinders)	7.5%

4 Transporting concrete

Methods: mortar pans, wheelbarrows, buckets and ropeways, truck mixers and dumpers, chutes, skips and hoists, belt conveyors, pumps and pipelines, and ready-mixed concrete.

Rule: bad handling causes **segregation**, so the method must keep the mix **cohesive** and must not need a change in workability.

Method	Use	Advantages and problems
Mortar pan	Small works	Labour-intensive. **No segregation**. In hot weather, large **water loss** from exposure

Method	Use	Advantages and problems
Wheelbarrow	At ground level, e.g. roads	Segregation on **rough roads**; reduced with **pneumatic tyres**
Buckets and ropeway	Valleys, high piers, long dam sites	Avoid **excessive free fall** of concrete (segregation)
Truck mixer and dumper	Long-lead concreting	Cover open trucks with **tarpaulin**; use **agitators** for long distances
Chute	Placing in **deep locations**	Slope **not flatter than 1V : 2.5H** or concrete will not slide. Do not change workability just to suit the chute. Not ideal technically, but widely used
Skip and hoist	**High-rise** structures	Skip travels vertically on rails, like a lift
Belt conveyor	Limited use	Segregation on steep slopes, at roller points and at changes of direction; over-exposure to the environment
Pump and pipeline	**Limited space**, or large pours **without cold joints**	Most sophisticated method (details below)

4.1 Pumping concrete

- Rates of **8 to 70 m^3 per hour**, up to about **300 m horizontally** and **90 m vertically**.
- Pipes **80–200 mm** in diameter, of steel, plastic or aluminium.
- Workability: minimum **slump 40–100 mm**, or **compacting factor 0.90–0.95**.
- Workability may drop by about **25%** by the delivery point (the concrete is compacted in the pipe), so allow for this in the mix design.

4.2 Ready-mixed concrete (RMC)

Concrete delivered for placing from a **central plant**.

Advantages:

- **Close quality control** of batching, so less variability.
- Useful on **congested sites** and in **highway construction**.
- **Agitator trucks** keep it from segregating and maintain workability.

- Convenient when **small quantities** or **intermittent placing** are needed.

Two types:

- **Central-mixed**: fully mixed at the plant; the truck only **agitates** it on the way.
- **Transit-mixed (truck-mixed)**: batched at the plant and **mixed in the truck**, either in transit or just before discharge.

The **speed of mixing** affects how fast the concrete stiffens, and the **number of revolutions** controls uniformity (up to about **300 revolutions** for mixing and agitating together).

4.3 Transit mixers

- Truck-mounted, capacity **4–7 m^3**; the most popular way to carry concrete over long distances, especially from RMC plants.
- Either the mixed concrete is **agitated at 2–6 rpm** all the way, or it is **batched at the plant and mixed in the truck** in transit or just before discharge.

4.4 Factors affecting the choice of transport

- Location of the site and **distance from the batching plant**
- Ease of access to the site
- Season and **weather**
- Size of the site and number of workers
- Location of the pour (height, depth)
- **Topography** of the site
- Cost

5 Placing concrete

Placing and compaction go on almost **simultaneously** and depend on each other. Together they decide the **strength, impermeability and durability** of the hardened concrete.

Main objective of placing: deposit the concrete **as close as possible to its final position**, so it does not **segregate** and can be **fully compacted**.

5.1 Rules for placing

1. **Do not move concrete sideways** by hand shovelling or with **poker vibrators** (this causes segregation).
2. Place concrete in **uniform layers**, not in large heaps or sloping layers.
3. Make each **layer thickness suit the vibration method**, so entrapped air can be removed from the bottom of each layer.
4. The **rate of placing should equal the rate of compaction**.
5. For a good finish and uniform colour on columns and walls, fill the forms at **at least 2 m per hour**, without delays.
6. **Fully compact each layer** before placing the next, and place each new layer **while the one below is still plastic**, so the concrete is **monolithic** (no cold joints).
7. **Avoid collision** of concrete with formwork and reinforcement.
8. Place concrete in a **vertical plane**. In horizontal or sloping forms, place it **vertically against, not away from**, the concrete already placed.

ACI good practice (shown as correct and incorrect sketches):

- **On a sloping surface:** use a **baffle** at the end of the chute so the concrete drops vertically; letting it run freely off the chute separates the coarse aggregate.
- **At the end of a chute or conveyor:** drop the concrete **vertically** (through a drop pipe or baffle) into the **centre** of the pile, not at an angle, which throws coarse aggregate to one side.
- **From buggies:** dump each load **against the face of concrete already placed**, not away from it, so the mortar and stone stay together.

5.2 Special placing techniques

Slip-forming

- A **continuous** process of placing and compacting concrete in a **moving form**.
- Uses **low-workability** concrete with **carefully controlled proportions** and **quick-setting** properties, so the concrete can support itself as the form moves on.

- Both **horizontal** (e.g. pavements, kerbs) and **vertical** slip-forming are possible. Vertical slip-forming is slower: the form is raised slowly (by jacks or crane) only when the concrete below has gained enough strength to support itself and the form.
- Set-up: a working **platform**, the **slip form** raised slowly as the concrete stiffens, **reinforcement** added by workers as it rises, **wet concrete** pumped in at the top, and stiff concrete **emerging at the bottom**.
- **High capital cost**, more than offset by the **very high rate of production**.
- Used for **bridges (piers), tall buildings (cores), towers, silos and dams**.

Shotcrete

- **"Shoot" + "concrete"**: concrete or mortar conveyed through a hose and **pneumatically projected at high velocity** onto a surface. The impact compacts it, so no formwork is needed on that face.
- Two processes:
  - **Dry-mix**: cement and **damp aggregate** are mixed and fed into a mechanical feeder (gun). Compressed air carries the material through the hose; **water is added at the nozzle**. (Equipment: upper feed chamber, lower working chamber, agitator, rotating feed wheel, nozzle.)
  - **Wet-mix**: **all ingredients including water are pre-mixed**, then pumped and sprayed; air is added at the nozzle.
- Uses: tunnel linings, slope stabilization, **repair and strengthening** of structures such as bridges, swimming pools, thin shells.

	Dry-mix	Wet-mix
Water added	At the nozzle (operator controls it)	Already in the mix
Rebound and dust	More	Less
w/c control	Depends on nozzle operator	Accurate
Output	Lower	Higher

Tremie

- Used for **deep forms** where normal compaction is impossible, and for **placing concrete under water**.
- **High-workability** concrete flows **by gravity** down a **vertical pipe** (the tremie). The pipe's lower end is kept **buried in the fresh concrete**, so the new concrete does not fall through water and wash out.
- The mix must be **cohesive**, without segregation or bleeding: **high cement content**, a **high proportion of fines**, and a workability aid such as a **pozzolan or admixture** (anti-washout VMA).

6 Compaction

Compaction expels the **entrapped air** from freshly placed concrete and packs the aggregate particles together, **increasing the density**.

Purpose:

- improve **strength and durability**,
- minimize **voids and permeability**,
- achieve **high density**, and
- eliminate **honeycombing** and other defects.

Entrapped air can be 5–20% of the volume of uncompacted concrete; each 1% of air left can lower strength by about 5–6%. **Low-slump concrete contains more entrapped air** than high-slump concrete, so it needs **more compaction effort**.

6.1 Choosing the method

Compaction can be manual or mechanical. The choice depends on:

- **amount and spacing of reinforcement**,
- **consistency** of the concrete, and
- **complexity of the formwork**.

6.2 Manual compaction

Used for **ordinary, small or less important** work, and where reinforcement is too congested for vibrators. The workability must be high enough to avoid honeycombing.

Method	How	Use
Rodding	Poking continuously over the whole area with a **2 m long, 16 mm diameter rod**, to pack the concrete and drive out air. Layers **15–20 cm** thick	General small work
Ramming	Pounding the surface	**Plain concrete at ground level** and around foundations; **not for RCC** or upper floors
Tamping	Beating the top surface with a **wooden cross-beam (10 × 10 cm)**; compacts and levels together	**Roof slabs and road pavements**

6.3 Mechanical compaction (vibration)

When concrete is vibrated, the **internal friction between aggregate particles is temporarily destroyed** and the concrete behaves like a **liquid**. It settles under gravity and **air bubbles rise** to the surface. Friction returns as soon as vibration stops. This is the modern, standard method.

Internal (needle or poker) vibrators

- The most common method for **cast-in-place** concrete (except where reinforcement is very congested).
- Parts: **motor**, **flexible shaft**, **vibrating head (poker)**. Inside the head an **eccentric (unbalanced) weight** spins at high speed, so the head vibrates in a circular motion.
- Head diameter **20–180 mm**. (The slide gives the length as 25–90 mm; heads are typically 250–900 mm long.)
- Powered by electricity, petrol or compressed air.
- Good practice: insert vertically at regular spacing so the zones of influence overlap, let it sink into the layer below, withdraw slowly; do not use it to push

concrete sideways.

External vibrators

- **Clamped rigidly to the formwork**, so the form and the concrete vibrate together.
- Forms must be **strong and well tied** to prevent distortion and **grout leakage**; considerable energy goes into vibrating the forms.
- Suitable for **thin members** and **precast units**.
- Frequency **1000–5000 cycles per minute**; gives a **good finish** but **uses more power** (less efficient than internal vibrators).
- Two kinds: vibrators attached to the formwork (**form vibrators, vibratory screeds, rammers**), and formwork attached to the vibrator (**vibrating tables, shock tables**).

Form vibrators

Clamped to the formwork; used for **very thin members** congested with reinforcement, to **supplement internal vibration**, and for **stiff mixes** where pokers cannot be used.

Vibrating tables

The mould is **clamped to the vibrating table**, which can vibrate horizontally and vertically. Used for **small quantities**: **laboratories** and some **precast** elements.

Surface (screed board) vibrators

- The vibrator is mounted on a **screed board** and vibrates the surface; similar in action to tamping.
- For **roof slabs and road pavements**; **not effective beyond about 15 cm depth**.
- **Vibratory screeds** for unreinforced slabs; **vibrating rollers** for thin slabs (and roller-compacted concrete); an **electric hammer** for compacting test cubes.

7 Defects from improper compaction

Inadequate vibration can cause:

- **Honeycombing**: voids between coarse aggregate particles where mortar has not filled them
- **Excessive entrapped air voids**
- **Sand streaks**
- **Cold joints**: a weak plane where a new layer was placed after the previous one had set
- **Placement lines**
- **Subsidence cracking**: cracks over reinforcement or at changes of section as the concrete settles

7.1 Defects of under-vibration

- **Excessive entrapped air voids**: under-vibration leaves a lot of air, which **reduces strength**.
- **Sand streaks**: heavy **bleeding washes the mortar out** along the form face, leaving streaks of sand; also caused by a **harsh mix** lacking fine aggregate.
- **Placement lines**: dark lines between adjacent layers, because the vibrator **did not penetrate into the layer below**.

7.2 Defects of over-vibration

- **Segregation**: heavier coarse aggregate settles and lighter material rises.
- **Bleeding**: water rises to the surface.
- **Formwork damage**.

Under-vibration is a more common problem than over-vibration.

Quick revision

- Production: batching → mixing → transporting → placing → compaction → curing.
- Weigh batching required by IS 456. Tolerances: cement ±1%, water ±1%, aggregate ±2%, admixture ±3%.

- Mixers: tilting drum (fast, unsegregated; low workability, large aggregate), non-tilting (slow, some segregation), pan/forced-action (stiff mixes, precast, labs), dual drum (highways, double output), continuous (dams, bridges).
- ASTM C94 uniformity: density 16 kg/m^3, air 1%, slump 25/40 mm, CA 6%, mortar density 1.6%, strength 7.5%.
- Chute slope ≥ 1V : 2.5H. Pump: 8–70 m^3/h, 300 m horizontal, 90 m vertical, pipe 80–200 mm, slump 40–100 mm. Transit mixer 4–7 m^3, agitate 2–6 rpm, ≤ 300 revolutions.
- Placing rules: near final position, uniform layers, place against previous concrete, new layer while old is plastic, ≥ 2 m/h in walls and columns.
- Slip-form (continuous, low workability, fast setting), shotcrete (dry-mix: water at nozzle; wet-mix: premixed), tremie (under water; pipe end buried; cohesive, rich mix).
- Manual compaction: rodding (2 m × 16 mm rod, 15–20 cm layers), ramming (PCC at ground), tamping (10 × 10 cm beam; slabs, roads).
- Vibrators: internal/poker (20–180 mm heads), external/form (1000–5000 cpm), vibrating table (precast, lab), surface/screed (≤ 15 cm).
- Under-vibration: air voids, honeycombing, sand streaks, placement lines, cold joints. Over-vibration: segregation, bleeding, form damage.

Questions to practise

1. What is batching? Compare volume and weigh batching and give the batching tolerances. (5 marks)
2. Describe the components of a batching plant. (5 marks)
3. Describe the different types of batch mixers with their advantages and uses. (8 marks)
4. How is the uniformity of mixing checked? (3 marks)
5. Describe the methods of transporting concrete, with the precautions for each. (10 marks)

6. Write short notes on: (a) pumping of concrete, (b) ready-mixed concrete, (c) transit mixers. (9 marks)
7. State the rules to be followed when placing concrete. (6 marks)
8. Write short notes on slip-forming, shotcrete and tremie concreting. (9 marks)
9. What is compaction? Why is it necessary? Describe the manual and mechanical methods of compaction. (10 marks)
10. What defects result from under-vibration and over-vibration? (5 marks)